

Hierarchical Clustering

What is Hierarchical Clustering?

Hierarchical Clustering is an **unsupervised machine learning algorithm** used to group similar data points into clusters.

Unlike K-Means:

- It does **not require K value initially**
- It creates a **tree-like structure** called a **Dendrogram**
- It works using **distance/similarity between points**

Why Do We Need Hierarchical Clustering?

K-Means has some limitations:

Problems in K-Means

1. Fails on Non-Spherical Data

K-Means works well only on circular/spherical clusters.

Example:

- Moon-shaped data
- Nested circles
- Irregular clusters

K-Means cannot separate them correctly.

2. Sensitive to Outliers

K-Means assigns every point to a cluster.

Even noisy points get assigned.

3. Requires K Before Training

You must manually tell:

- number of clusters (K)

Sometimes we do not know the correct value.

Main Idea of Hierarchical Clustering

Instead of directly creating K clusters:

It creates a hierarchy (levels) of clusters.

Like:

- Small groups merge
- Then larger groups form
- Until all points become one cluster

Types of Hierarchical Clustering

1. Agglomerative Clustering (Bottom-Up)

Most commonly used.

Working:

- Start with every point as separate cluster
- Merge closest clusters repeatedly
- Continue until one cluster remains

Example:

Initially:

A B C D

After merging:

(A, B) (C, D)

Then:

$((A, B), (C, D))$

2. Divisive Clustering (Top-Down)

Opposite approach.

Working:

- Start with one large cluster
- Keep dividing into smaller clusters

Less commonly used.

Agglomerative Clustering Working

Suppose we have points:

P1 P2 P3 P4 P5

Step 1:

Every point becomes its own cluster.

[P1] [P2] [P3] [P4] [P5]

Step 2:

Find two closest points.

Example:

P1 and P2

Merge them:

[P1, P2]

Step 3:

Again find closest clusters.

Merge repeatedly.

Final Step:

All points become one cluster.

Dendrogram

A Dendrogram is a tree diagram showing:

- Which clusters merged
- At what distance they merged

Importance

Using dendrogram we can decide:

- Optimal number of clusters

Choosing Number of Clusters using Dendrogram

Rule:

Cut dendrogram from the longest vertical gap.

The number of lines cut = number of clusters.

Example:

If horizontal cut crosses 3 vertical branches:

Then:

Number of clusters = 3

Distance Metrics in Hierarchical Clustering

Different methods calculate distance differently.

1. Single Linkage (Minimum Distance)

Distance between two clusters =
minimum distance between points.

Formula intuition:

Closest point to closest point

Advantage

- Works well for connected shapes

Disadvantage

- Sensitive to noise/outliers

2. Complete Linkage (Maximum Distance)

Distance between clusters =
maximum distance between points.

Formula intuition:

Farthest point to farthest point

Advantage

- Handles outliers better

Disadvantage

- Can break large clusters

3. Average Linkage

Distance =
average of all pair distances.

Balanced approach.

4. Ward Linkage

Most popular method.

It minimizes cluster variance.

Goal:

- Keep clusters compact
- Reduce internal variance

Important Parameters

1. n_clusters

Number of clusters required.

2. linkage

Method used:

- single
- complete
- average
- ward

3. metric / affinity

Distance formula:

- Euclidean
- Manhattan
- Cosine

Advantages of Hierarchical Clustering

1. No Need for Initial K

Unlike K-Means.

2. Dendrogram Gives Visualization

Easy to understand cluster relationships.

3. Works on Different Cluster Shapes

More flexible than K-Means.

Disadvantages

1. Slow for Large Datasets

Time complexity is high.

2. Memory Expensive

Requires distance matrix.

If dataset is huge:

- RAM usage becomes large.

3. Difficult to Revert

Once clusters merge:

- Cannot undo.

Difference Between K-Means and Hierarchical Clustering

Feature	K-Means	Hierarchical
Need K initially	Yes	No
Uses centroid	Yes	No
Dendrogram	No	Yes
Handles irregular shapes	Poor	Better
Speed	Faster	Slower
Large datasets	Better	Poor

Real World Applications

Customer Segmentation

Grouping customers by:

- income
- spending
- behavior

Document Clustering

Grouping similar documents/articles.

Biology

Gene analysis and species classification.

Image Segmentation

Grouping similar image regions.